SUMMARY

(1) Equation of the circle in a standard form is $(x-h)^2+(y-k)^2=r^2$.

(i) Centre (h,k) (ii) radius 'r'

(2) Equation of a circle in general form is $x^2+y^2+2gx+2fy+c=0$.

(i) centre $(-g,-f)$ (ii) radius = $\sqrt{g^2+f^2-c}$

(3) The circle through the intersection of the line $lx+my+n=0$ and the circle $x^2+y^2+2gx+2fy+c=0$ is $x^2+y^2+2gx+2fy+c+\lambda(lx+my+n)=0, \lambda \in \mathbb{R}^1$.

(4) Equation of a circle with (x_1,y_1) and (x_2,y_2) as extremities of one of the diameters is $(x-x_1)(x-x_2)+(y-y_1)(y-y_2)=0$.

(5) Equation of tangent at (x_1,y_1) on circle $x^2+y^2+2gx+2fy+c=0$ is $xx_1+yy_1+g(x+x_1)+f(y+y_1)+c=0$

(6) Equation of normal at (x_1,y_1) on circle $x^2+y^2+2gx+2fy+c=0$ is $yx_1-xy_1+g(y-y_1)-f(x-x_1)=0$.

Table 1

Tangent and normal

Curve	Equation	Equation of tangent	Equation of normal
Circle	$x^2+y^2=a^2$	(i) cartesian form $xx_1+yy_1=a^2$ (ii) parametric form $x\cos\theta+y\sin\theta=a$	(i) cartesian form $xy_1-yx_1=0$ (ii) parametric form $x\sin\theta-y\cos\theta=0$
Parabola	$y^2=4ax$	(i) $yy_1=2a(x+x_1)$ (ii) $yt=x+at^2$	(i) $xy_1+2y=2ay_1+x_1y_1$ (ii) $y+xt=at^3+2at$
Ellipse	$\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$	(i) $\frac{xx_1}{a^2}+\frac{yy_1}{b^2}=1$ (ii) $\frac{x\cos\theta}{a}+\frac{y\sin\theta}{b}=1$	(i) $\frac{a^2x}{x_1}+\frac{b^2y}{y_1}=a^2-b^2$ (ii) $\frac{ax}{\cos\theta}-\frac{by}{\sin\theta}=a^2-b^2$
Hyperbola	$\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$	(i) $\frac{xx_1}{a^2}-\frac{yy_1}{b^2}=1$ (ii) $\frac{x\sec\theta}{a}-\frac{y\tan\theta}{b}=1$	(i) $\frac{a^2x}{x_1}+\frac{b^2y}{y_1}=a^2+b^2$ (ii) $\frac{ax}{\sec\theta}+\frac{by}{\tan\theta}=a^2+b^2$

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